

The Passive Layer:

A Physically Necessary but Uncalculated Term
in Galactic Dynamics

B. Sun

Seoul Inside

seoulinside.substack.com

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Abstract

We identify a term absent from all standard cosmological frameworks: the propagating gravitational influence of mass that has ceased to exist. This term — the *Passive Layer*, also called *Ghost Mass* — is a logical consequence of two independently confirmed facts: (1) gravity propagates at the speed of light [GW170817; Abbott et al., 2017], and (2) mass is continuously created and destroyed throughout the universe (pair production and annihilation; foundational to quantum field theory). Combined, these facts entail that the gravitational signal of annihilated mass continues to propagate after the source has gone. This effect does not cancel: gravity has no negative mass. It reaches dynamic equilibrium through cosmic expansion, analogous to Olbers' paradox. The Passive Layer comprises three independent physical components: (A) vacuum fluctuations, (B) local astrophysical pair production near black holes, and (C) retarded gravitational interaction from black hole displacement. This document does *not* claim that the Passive Layer replaces dark matter. It claims something narrower: a physically necessary, non-zero term has been omitted from standard models, and its magnitude has never been calculated. Companion simulations (GALAXYCS v4, YANG-MILLS COLLIDER v3.2) demonstrate that Component C alone reproduces flat rotation curves and spontaneous spiral arm formation without free parameters. This research originated independently, without prior knowledge of the dark matter literature or of Yahalom [2013]'s work on retarded gravity, constituting an independent derivation of the same physical effect from an orthogonal direction.

1. Introduction: An Independent Discovery

This research did not begin as an attempt to explain dark matter. At the time the core observations were made, the author had no prior familiarity with the dark matter literature or with existing work on retarded gravity in galactic dynamics [Yahalom, 2013, 2019, 2024].

The research began with the construction of a browser-based particle physics simulator — the *Yang-Mills Collider* — implementing relativistic particle motion, the Boris integrator, the Bethe-Bloch formula, QCD running coupling, and 4-momentum conservation. When a black hole was added to this simulator, an accretion disk appeared *spontaneously*. No accretion disk code had been written. The disk emerged from the interaction of Lorentz-force charge separation and Newtonian gravity alone.

This observation motivated a second simulator — GALAXYCS v4 — designed to investigate galactic dynamics with retarded gravitational propagation. In GALAXYCS v4, whenever the central black hole was displaced from the disk’s geometric centre by any non-zero amount, spiral arm structure emerged immediately and persisted indefinitely. When the black hole was held stationary, no spiral structure appeared.

The question then became: is black hole displacement a special condition, or a default state? Analysis showed it to be the default (Section 7). Only after these observations were made did comparison with the existing literature reveal that: (a) flat rotation curves had long been attributed to dark matter; (b) Yahalom [2013] had independently developed a framework for retarded gravity in galactic dynamics; and (c) fifty years of dark matter particle searches had produced null results.

The fact that this derivation was reached independently, from a simulation rather than from theory, constitutes a second confirmation of the underlying physical mechanism from an orthogonal direction.

2. Two Established Facts and Their Logical Consequence

2.1. Fact 1: Gravity Propagates at the Speed of Light

A change in a gravitational source does not instantly affect the surrounding universe; the influence travels outward at c . This was confirmed in 2017 when LIGO/Virgo detected gravitational waves and gamma rays from a neutron star merger (GW170817) arriving simultaneously, constraining the speed of gravity to within 10^{-15} of c [Abbott et al., 2017].

Consequence: If a mass disappears, its gravitational signal continues to travel outward for time r/c after the mass is gone.

2.2. Fact 2: Mass Is Created and Destroyed

Energy converts into mass and mass converts back into energy. Particle-antiparticle pairs are continuously created and annihilated throughout the universe — in vacuum fluctuations, near black holes, and in high-energy environments. This has been directly observed and is a foundational result of quantum field theory [Schwinger, 1951, Hawking, 1975, Unruh, 1976].

2.3. The Logical Consequence

Combine Fact 1 and Fact 2. A mass exists. It generates a gravitational signal propagating outward at c . The mass then ceases to exist. **The gravitational signal does not cease.** It continues to travel and continues to exert influence on everything it reaches.

This propagating remnant is the **Passive Layer** — also referred to as *Ghost Mass* or *gravitational reverberation*. Formally, the gravitational influence of a mass that existed at distance r persists for time

$$\tau(r) = \frac{r}{c} \tag{1}$$

after that mass is gone. This is not a hypothesis; it is a consequence of two confirmed facts.

3. Why the Passive Layer Does Not Cancel

Gravity has no negative mass. Particles and antiparticles carry opposite electric charges, but they carry the *same* gravitational mass. The gravitational reverberation of a particle and its antiparticle therefore point in the same direction. They do not cancel; they accumulate.

What appears to be cancellation in homogeneous regions is not cancellation — it is balance. Consider the Earth: every atom pulls on every other atom; none of these forces cancel; they sum. At the centre the net force is zero not because forces cancel, but because they balance symmetrically. Move away from the centre and the imbalance becomes immediately apparent.

The universe is not perfectly homogeneous anywhere. The net effect of the Passive Layer is therefore *non-zero everywhere*.

4. Dynamic Equilibrium

The Passive Layer does not accumulate without bound. Two opposing processes maintain equilibrium:

- **Generation:** New reverberations are continuously created via pair production.

- **Dilution:** Existing reverberations are diluted by cosmic expansion.

This is structurally identical to Olbers' Paradox. The night sky is dark not because stars do not emit light, but because the universe is finite in age and expanding. The Passive Layer reaches a stable equilibrium density for the same reasons:

$$\frac{d\rho_{\text{PL}}}{dt} = \dot{\rho}_{\text{gen}} - \dot{\rho}_{\text{dil}} = 0 \quad (\text{at equilibrium}). \quad (2)$$

The underlying process is extraordinarily dynamic, but the background density at any given moment is effectively constant.

5. Three Independent Physical Components

The Passive Layer is not a single monolithic effect. It comprises three independent physical mechanisms.

5.1. Component A: Vacuum Fluctuations (Universal, Spatially Constant)

This component originates from the quantum vacuum. In flat spacetime, vacuum fluctuations are spatially uniform — the same in every cubic Planck-length regardless of position:

$$\dot{M}_{\text{pair}}^{\text{vac}}(r) = \text{constant} \quad (r\text{-independent}). \quad (3)$$

Component A dominates the far-field regime ($r \gg R_{\text{disk}}$) and is the physical origin of the flat rotation curve derivation in Section 6.

5.2. Component B: Local Astrophysical Sources (Disk-Like)

This component originates from high-energy environments around black holes, AGN, supernovae, and magnetars. In the ultra-strong magnetic fields observed near supermassive black holes ($10^4\text{--}10^6$ G near the ISCO; [EHT MWL 2021](#)), particle-antiparticle pairs produced from vacuum fluctuations are separated by the Lorentz force:

$$\mathbf{F} = q(\mathbf{v} \times \mathbf{B}). \quad (4)$$

The Larmor radius

$$r_L = \frac{\gamma m v_{\perp}}{|q|B} \quad (5)$$

depends on particle mass; positive and negative charges curve in opposite directions. Black hole gravity pulls both toward the centre, but they arrive from opposite sides and do not annihilate. They persist as mass.

This behaviour was observed directly in YANG-MILLS COLLIDER v3.2: no accretion-disk code was written, yet a disk appeared spontaneously from Lorentz force and Newtonian gravity alone.

Component B scales with local energy density and dominates the near-field regime ($r \lesssim R_{\text{disk}}$).

5.3. Component C: Retarded Gravity and Black Hole Displacement

When a black hole is displaced from the geometric centre of a disk galaxy, stars at different distances reference different past positions of that black hole. The retarded position is

$$\mathbf{r}_{\text{BH}}^{\text{ret}}(t, r) = \mathbf{r}_{\text{BH}}\left(t - \frac{\alpha r}{c}\right), \quad (6)$$

where α is the retardation strength parameter (physical prediction: $\alpha = 1$). In a differentially rotating disk, this radially dependent angular offset is sheared into a spiral pattern.

The retarded acceleration term [Yahalom, 2013] is

$$\mathbf{a}_{\text{ret}} = \alpha \cdot \frac{G \ddot{M}_{\text{BH}}}{2c^2} \hat{R}. \quad (7)$$

Critically, unlike Newtonian gravity ($\propto r^{-2}$), this term does not diminish with distance. Retarded gravitational effects therefore become *relatively more important* at large radii — precisely where observed rotation curves deviate most strongly from Newtonian predictions.

The total pair production rate is the sum of both position-dependent components:

$$\dot{M}_{\text{pair}}^{\text{total}}(r) = \dot{M}_{\text{pair}}^{\text{vac}} + \dot{M}_{\text{pair}}^{\text{local}}(r). \quad (8)$$

6. The Ghost Mass Formula and Flat Rotation Curves

The accumulated ghost mass up to radius r is

$$M_{\text{ghost}}(r) = \int_0^r \dot{M}_{\text{pair}}(r') \frac{r'}{c} dr'. \quad (9)$$

When the pair production rate is spatially constant (Component A dominates):

$$M_{\text{ghost}}(r) = \dot{M}_{\text{pair}} \cdot \frac{r}{c} \implies M_{\text{ghost}} \propto r. \quad (10)$$

The consequence for rotation curves follows directly from centrifugal balance:

$$v_c^2(r) = \frac{G M_{\text{ghost}}(r)}{r} = \frac{G \dot{M}_{\text{pair}}}{c} = \text{constant}. \quad (11)$$

If $M_{\text{ghost}} \propto r$, rotation velocity becomes constant regardless of radius — the observed flat rotation curve — emerging without dark matter and without free parameters. The single quantity $\dot{M}_{\text{pair}}$ is determined by observational data (Section 8).

7. Black Hole Displacement Is the Default State

The spiral arm mechanism (Component C) requires black hole displacement. Whether this is a special condition or a default state is answerable from first principles.

In the Solar System, the Sun contains $\approx 99.8\%$ of total system mass; it cannot be meaningfully displaced. In a disk galaxy, the central supermassive black hole contains only 0.1–0.5% of total galactic mass [Kormendy & Ho, 2013], expressed as the mass fraction

$$f_{\text{BH}} = \frac{M_{\text{BH}}}{M_{\text{total}}} \approx 0.001\text{--}0.005. \quad (12)$$

The black hole has no structural reason to remain at the barycenter. Perfect stasis would require exact simultaneous cancellation of:

- host galaxy peculiar velocity (100–600 km s^{−1});
- tidal forces from satellite galaxies and globular clusters;
- galactic bar oscillations;
- large-scale structure gravitational background;
- recoil from asymmetric gravitational wave emission.

This exact cancellation is not physically plausible. **Black hole displacement is the default state.** Spiral structure is therefore the *expected* morphology for disk galaxies.

Observational support: Batcheldor et al. [2010] measured a 6.8 ± 0.8 pc projected displacement in M87; Bartlett et al. [2021] found 10–100 pc displacements to be the most commonly observed range; Chu et al. [2022] find that one-third of brightest cluster galaxy black holes are off-centre at $z = 0$, with offsets sustained for up to 6 Gyr.

8. The Parameter Problem Is Already Solved

The pair production rate $\dot{M}_{\text{pair}}$ is the only free parameter in Equation (10). It does not need to be calculated from first principles.

Decades of dark matter research have produced precise measurements of the ‘missing mass’ at every scale:

- galaxy rotation curves $\rightarrow$ required mass distribution $M(r)$;

- gravitational lensing $\rightarrow$ required mass density $\rho(r)$;
- CMB + BAO $\rightarrow \Omega_c h^2 = 0.118$ [Planck Collaboration, 2020];
- NFW profile parameters for individual halos.

These measurements are not constraints on dark matter alone. If the Passive Layer is the correct explanation, they are direct measurements of $\dot{M}_{\text{pair}}$. The existing observational dataset has already determined the value of every parameter in the theory.

Key methodological distinction. Conventional alternative theories must independently calculate their predicted magnitude before comparison to observation. The Passive Layer inverts the procedure: *observation supplies the answer; theory supplies the mechanism.*

9. Distinction from Yahalom (2013, 2019, 2024)

Yahalom [2013] developed a theoretical framework for retarded gravity in galactic dynamics that shares the use of delayed gravitational propagation. The distinction is fundamental:

Aspect	Yahalom	Passive Layer (Component C)
Source of effect	Existing mass with retarded propagation	Mass that has <i>ceased to exist</i>
Physical question	How does existing mass gravitate with delay?	What happens to gravity after the source mass is gone?
Mass budget	Current mass distribution	Cumulative history of pair production and annihilation
Derivation origin	Theoretical (GR linearisation)	Observational (simulation $\rightarrow$ emergent behaviour)

Yahalom asks how the gravity of *existing* mass propagates with a delay. The Passive Layer asks what happens to gravity after the source mass *is gone*. These are different questions; they may both be correct simultaneously. The Passive Layer does not supersede Yahalom’s work — it identifies an additional, independent term that Yahalom’s framework does not address.

It is noted that this distinction was not apparent to the author during the original derivation, as Yahalom’s work was unknown at the time. The convergence on similar physical mechanisms from independent directions is consistent with the reality of the underlying effect.

10. Simulation Evidence

10.1. GalaxyCS v4 — Spiral Arm Formation

GALAXYCS v4 is an N-body simulator implementing retarded gravitational propagation via a FIFO position-history buffer. With 20,000–80,000 test particles initialised on circular orbits:

Condition	Result
BH at rest (displacement = 0)	No spiral structure; axisymmetric disk
BH displaced by any non-zero amount	Spiral arm structure emerges immediately and persists indefinitely
Retarded gravity disabled	Keplerian rotation curve ($v \propto r^{-1/2}$)
Retarded gravity enabled + BH displacement	Flat rotation curve + spontaneous spiral arms

The transition is not gradual. There is no critical displacement threshold. Any non-zero displacement produces spiral structure. The mechanism is geometric, not dynamic; it requires no special initial conditions, no resonance, and no tuned parameters.

Arm morphology is determined by the kinematic history of BH displacement: unidirectional displacement produces two-arm spirals; oscillatory displacement produces multi-arm structure; irregular displacement produces flocculent or asymmetric arms. This range of morphologies is consistent with the observed diversity of spiral galaxy types.

10.2. Yang-Mills Collider v3.2 — Spontaneous Accretion Disk

YANG-MILLS COLLIDER v3.2 implements relativistic motion (Lorentz factor), the Boris integrator [standard in GEANT4 and plasma PIC codes; [Boris, 1970](#)], Bethe-Bloch energy loss, 4-momentum conservation with Lorentz boost, QCD running coupling $\alpha_s(\mu)$, and Kerr black hole gravity.

When a black hole was introduced into a high-energy particle collision environment, an accretion disk appeared without being programmed. The Lorentz force separated particle-antiparticle pairs; black hole gravity confined them; annihilation was suppressed; mass persisted.

“I did not build an accretion disk. One appeared.”
— *I Added a Black Hole to the LHC* (2026-06-06)

This constitutes direct simulation evidence for Component B: local astrophysical pair separation and mass persistence near black holes.

11. Comparison with Existing Frameworks

Aspect	Λ CDM	MOND	Yahalom	Passive Layer
Dark matter	Particle (WIMP/axion)	Not required	Reduced	Uncalculated term
Rotation curves	Halo fitting	Excellent fit	Retarded dy- namics	$M_{\text{ghost}} \propto r$
Spiral arms	Not addressed	Not addressed	Not addressed	BH displace- ment
Bullet Cluster	Non-collisional DM	Cannot explain	Not addressed	Non-collisional reverberation
Direct detec- tion	Possible in principle	N/A	N/A	Impossible in principle
Free parame- ters	Multiple	a_0	α	None (from ob- servation)
CMB fit	Excellent (0.1%)	Not addressed	Not addressed	Untested

12. The One Open Question

The *existence* of the Passive Layer is a logical consequence of confirmed physics. It is not in question. The one question that remains open is: **How large is it?**

The following open problems are explicitly acknowledged:

- Q1. Cancellation mechanism.** The 10^{123} vacuum energy cancellation mechanism is unknown. Without it, quantitative estimates of Component A remain unconstrained. *Status: open problem in quantum gravity.*
- Q2. Backreaction.** Self-consistent treatment requires coupled QFT-in-curved-spacetime and Einstein field equations. *Status: unsolved even in simplified toy models.*
- Q3. Distinguishability from CDM.** Near galactic centres, the Passive Layer profile and the CDM profile may produce similar observational signatures. Requires equation-of-state measurements or high-precision inner (< 1 kpc) rotation curve data. *Status: feasible but not yet performed systematically.*
- Q4. Cosmological consistency.** The integrated contribution over all galaxies in the observable universe may affect the CMB power spectrum and BAO scale. *Status: not yet calculated.*

Q5. N-body resolution. GALAXYCS v4 qualitative behaviour has not been verified at higher particle counts. *Status: requires higher-resolution simulation.*

Q6. Systematic rotation curve fitting. Qualitative agreement with M33 has been observed. A systematic fit to the SPARC database [175 galaxies; [Lelli et al., 2016](#)] has not been performed. *Status: required next step.*

13. Falsifiable Predictions

Prediction	Method	Status
BH displacement correlates with spiral arm strength	SMBH offset measurements vs. arm pitch angle	Testable now
Galaxies with centred BHs have no spiral structure	High-resolution imaging + astrometry	Testable now
Rotation curve excess scales with galaxy age (mass-conversion history)	Age-matched rotation curve comparison	Requires large survey
No dark matter particle detected	Direct detection experiments	Null results consistent
Void formation from BH mass fluctuation	Large-scale structure vs. AGN activity	Feasible

14. Conclusion

We have identified and characterised a physically necessary term absent from all standard cosmological frameworks: the propagating gravitational influence of mass that has ceased to exist.

This term — the Passive Layer — follows necessarily from two independently confirmed facts. It does not cancel. It reaches dynamic equilibrium. It comprises three independent physical components. Its magnitude reproduces the observational signature of dark matter without requiring a new particle.

The Passive Layer does not claim to replace dark matter. It claims something more fundamental: a term exists, it is non-zero, and it has never been calculated. Whether its magnitude accounts for all, some, or a negligible fraction of the observed missing mass is an open question — but one that can be answered with existing observational data and existing theoretical tools.

*The universe has been accounting for the Passive Layer since the beginning.
Our models have not.*

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A. Simulation Infrastructure

GalaxyCS v4

- N-body galaxy simulator with retarded gravitational propagation
- FIFO history buffer for black hole position; linear interpolation for retarded position retrieval
- Leapfrog (velocity Verlet) integrator: $\mathcal{O}(\Delta t^2)$ accuracy, symplectic, time-reversible
- Real-time HUD: rotation velocity, Newtonian prediction, ratio, energy and angular momentum conservation monitoring
- Test particle count: 20,000–80,000; black hole displacement: continuously adjustable

Yang-Mills Collider v3.2

- Relativistic motion: Lorentz factor $\gamma = E/m$; speed of light enforced
- Boris integrator: standard algorithm used in GEANT4 and plasma PIC codes
- Bethe-Bloch energy loss: particle mass- and charge-dependent
- 4-momentum conservation: CM frame 2-body decay with Lorentz boost to lab frame
- Branching ratios: PDG values ($\pi^+ \rightarrow \mu^+ \nu_\mu$: 99.99%; $\pi^+ \rightarrow e^+ \nu_e$: 0.01%)

- QCD running coupling: 1-loop β function with quark flavour thresholds
- Kerr black hole gravity: implemented as Newtonian approximation with mass parameter
- Magnetic field: solenoid configuration, adjustable strength